THE PROJECTION FORMULA FOR A CLOSED IMMERSION OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement involves a morphism of ringed topoi induced by the closed immersion; that construction is part of the statement span and the proof uses only the two adjunction maps written there.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Exactness of the stalk functor on the etale site of an algebraic space. Properties of Spaces, Theorem spaces-properties-theorem-exactness-stalks, used at proof lines 2672-2674. Reduces the isomorphism of sheaves to an isomorphism on stalks.
- Behaviour of stalks under pullback of sheaves. Properties of Spaces, Lemma spaces-properties-lemma-stalk-pullback, used at proof lines 2688-2691. Identifies the stalks of the pullbacks appearing in the computation.
- Behaviour of stalks under pushforward along a closed immersion. Lemma lemma-stalk-push-closed, used at proof lines 2691-2693. Identifies the stalk of f_*G at xbar with the stalk of G at zbar.
- The projection formula as an adjunction between pullback and pushforward. used at proof lines 2662-2671. Constructs the canonical map whose isomorphism is asserted.

Source reference. Original lemma 0DK1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16996–16996. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04K2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5958–5958. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original theorem 04K5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5961–5961. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $i : Z \to X$ be a closed immersion of algebraic spaces over S. Let $\mathcal{A}$ be a sheaf of rings on $X_{étale}$. Let $\mathcal{B}$ be a sheaf of rings on $Z_{étale}$. Let $\varphi : \mathcal{A} \to i_{small,*}\mathcal{B}$ be a homomorphism of sheaves of rings so that we obtain a morphism of ringed topoi*

$$f : (Sh(Z_{étale}), \mathcal{B}) \longrightarrow (Sh(X_{étale}), \mathcal{A}).$$

For a sheaf of $\mathcal{A}$-modules $\mathcal{F}$ and a sheaf of $\mathcal{B}$-modules $\mathcal{G}$ the canonical map

$$\mathcal{F} \otimes_{\mathcal{A}} f_*\mathcal{G} \longrightarrow f_*(f^*\mathcal{F} \otimes_{\mathcal{B}} \mathcal{G}).$$

is an isomorphism.

Proof. The map is the map adjoint to the map

$$f^*\mathcal{F} \otimes_{\mathcal{B}} f^*f_*\mathcal{G} = f^*(\mathcal{F} \otimes_{\mathcal{A}} f_*\mathcal{G}) \longrightarrow f^*\mathcal{F} \otimes_{\mathcal{B}} \mathcal{G}$$

coming from $\mathrm{id} : f^*\mathcal{F} \to f^*\mathcal{F}$ and the adjunction map $f^*f_*\mathcal{G} \to \mathcal{G}$. To see this map is an isomorphism, we may check on stalks (Properties of Spaces, Theorem 04K5). Let $\overline{z} : \mathrm{Spec}(k) \to Z$ be a geometric point with image $\overline{x} = i \circ \overline{z} : \mathrm{Spec}(k) \to X$. Working out what our maps does on stalks, we see that we have to show

$$\mathcal{F}_{\overline{x}} \otimes_{\mathcal{A}_{\overline{x}}} \mathcal{G}_{\overline{z}} = (\mathcal{F}_{\overline{x}} \otimes_{\mathcal{A}_{\overline{x}}} \mathcal{B}_{\overline{z}}) \otimes_{\mathcal{B}_{\overline{z}}} \mathcal{G}_{\overline{z}}$$

which holds true. Here we have used that taking tensor products commutes with taking stalks, the behaviour of stalks under pullback Properties of Spaces, Lemma 04K2, and the behaviour of stalks under pushforward along a closed immersion Lemma 0DK1. □